Fiscal Year:	FY 2026	**Task Last Updated:**	FY 03/04/2026
PI Name:	Bouxsein, Mary Ph.D.		
Project Title:	Dose-Response Study of Musculoskeletal Outcomes Following Centrifugation in Adult Mice on ISS		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Muscle**:Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance (3) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mbouxsei@bidmc.harvard.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-667-4594
Organization Name:	Beth Israel Deaconess Medical Center/Harvard Medical School		
PI Address 1:	Department of Orthopedic Surgery		
PI Address 2:	330 Brookline Ave, RN115		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	02/14/2019	**End Date:**	06/01/2026
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 06/01/2026 per S. Mack-Phillips/JSC (Ed., 3/24/25) NOTE: End date changed to 06/01/2025 per S. Mack-Phillips/JSC (Ed., 2/19/25) NOTE: End date changed to 06/01/2024 per NSSC information (Ed., 11/25/22) NOTE: End date changed to 12/31/2022 per NSSC information (Ed., 2/8/22) NOTE: End date changed to 12/31/2021 per NSSC information (Ed., 4/7/21) NOTE: End date changed to 12/15/2020 per NSSC information (Ed., 7/24/20)20)		

Key Personnel Changes/Previous PI:	June 2025 report: The PI has changed the list of Co-Investigators. Virginia Ferguson, Ph.D. is no longer on the list of Co-Investigators (Ed., 6/12/25). Additional note, per the PI: Co-Investigator Charles Farber, Ph.D. has left the project (Ed., 2/15/23). November 2020 report: Marc Wein, MD, PhD has been added as a co-investigator due to his expertise in mechanobiology and multi-omic analyses.
COI Name (Institution):	Rutkove, Seward M.D. (Beth Israel Deaconess Medical Center, Inc./Harvard Medical School) Willey, Jeffrey Ph.D. (Wake Forest University) Wein, Marc M.D., Ph.D. (Masachusetts General Hospital)
Grant/Contract No.:	80NSSC19K0534
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Mechanical loading is required for maintenance of the musculoskeletal system. Thus, exposure to spaceflight or reduced mechanical loading on Earth induces marked bone loss, muscle atrophy, and degradation of soft-tissue structures in both the knee (e.g., cartilage, menisci, and ligaments) and hip (e.g., cartilage) joints. This is a major concern for astronauts during and after long-duration spaceflight, as they may be at increased risk for reduced performance, bone fractures, and both early-onset osteoporosis and arthritis. Artificial gravity, generated by centrifugal force generation, is a possible approach to mitigate these deleterious changes. Yet, the ability of partial gravity induced by centrifugal acceleration to inhibit adverse musculoskeletal changes in spaceflight remains unknown. Given the constraints of studying centrifugation as a countermeasure on Earth, spaceflight-based studies are needed. We propose to determine the effects of varying partial gravity levels on bone, muscle, and soft tissues of the hip and knee joints in adult mice flown aboard the International Space Station (ISS) in the Japan Aerospace Exploration Agency (JAXA) Mouse Habitat Unit. We will examine bone structure post-flight using high-resolution microcomputed tomography (microCT); bone cellularity using quantitative histomorphometry; bone function via biomechanical testing; and bone composition via Raman spectroscopy and quantitative backscattered electron imaging. We will examine neuromuscular function via pre- and post-flight gait analysis, balance beam walking, and grip strength measurements. Post-flight muscle analyses will include histology and electrical impedance myography. Post-flight analyses of joint soft-tissues will include structural measurements of cartilage, menisci, and ligaments using both contrast-enhanced high-resolution microCT and histology; molecular composition of cartilage and menisci using proteomics and Raman spectroscopy; and biomechanical properties of cartilage using nano-indentation. Cellular and molecular responses for bone, muscle, and joint soft tissues will also be evaluated via whole transcriptome analyses (e.g., RNASeq). Results from these integrated, comprehensive analyses will provide information regarding whether partial gravity, either induced by centrifugal acceleration or via Moon or Mars environments, will protect from musculoskeletal deterioration during spaceflight, or whether additional countermeasures will be necessary.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A better understanding of the effects of different levels of mechanical loading via centrifugation on the musculoskeletal system may inform interventions and rehabilitation protocols for individuals exposed to chronic immobilization or unloading.
	Artificial gravity (AG) by centrifugation is a potential countermeasure to mitigate deleterious musculoskeletal changes associated with microgravity (µG). Prior work showed that exposure to 1G by artificial gravity inhibited musculoskeletal deterioration associated with spaceflight (Shiba et al., 2017). Yet, the responses of the musculoskeletal system to varied magnitudes of AG are not known. To address this knowledge gap, an interdisciplinary project team of investigators from the US and Japan designed an integrated protocol to assess the effects of varied levels of artificial gravity on various physiological systems. Our team focused on the musculoskeletal system. Adult (12 weeks) male C57Bl/6J mice were exposed to either microgravity alone (0G), or to 0.33G, 0.67G, or 1G via centrifugation using the JAXA Multiple Artificial-gravity Research System (MARS) during the Joint Partial-Gravity Rodent Research (JPG-RR)/Mouse Habitat Unit 8 (MHU-8) 30-day mission on the International Space Station (ISS) (n=5-6/group). Ground controls were housed in an identical environment (HGC, n=12). The launch occurred in March 2023, with a successful live animal return in April 2023. Before launch and after the live animal return, we assessed gait performance, grip strength, bone mass, and body composition. Bone and muscle tissues were harvested at necropsy for morphologic and transcriptomic analyses (via bulk RNASeq). In addition, we used electrical impedance myography to assess the intrinsic muscle properties of the gastrocnemius. In addition, we collected blood cell-depleted femoral bone marrow (n=5-6/group) postflight in the 0G and 1G groups for single-cell RNA sequencing using the HIVE platform (Honeycomb). Preliminary results are presented below. Gait analyses: All pre-flight gait parameters for fore- and hind limbs were similar between groups. Gait parameters for GC, VIV, and 1g were unchanged pre- to post-flight. Mice exposed to 0G failed to perform post-flight, unable to locomote and frequently rolling onto their backs: µg was excluded from quantitative analysis. Moreover, 50% of mice from the 0.33G group had similar performance failures, limiting the testable group size (n=3). All other mice (0.67G and 1G) performed post-flight. In general, changes in gait parameters were altered in a gravity-dependent manner, with post-flight deficits in the 0.33G group greatest vs VIV, GC, and 1g (Fig. 1); 0.67g attenuated the deleterious changes vs 0.33g. The magnitude of several gait pattern changes between pre- and post-flight in hind limbs were inversely related to gravitational loading, including: Δswing time (reflecting ataxia); Δswing/stride ratio (neuromotor deficits); stride width variability (higher risk of falling); and others consistent with RR9 (e.g., stride width). Moreover, changes in stance time (motor abnormalities) and propel time (consistent with RR9) were directly related to gravitational loading. Several parameters (e.g., stride frequency and stride length) were unchanged, though were limited in number. Forelimb patterns resembled hind limbs. Gait pattern changes in mice reflective of neuromotor deficits and instability, among others, occurred in a gravity-dependent manner after spaceflight. Continuous exposure to artificial gravity at 1G prevented these deficits; 0.67G attenuated these deficits. Exposure to 0.33G (Mars gravity) resulted in pronounced gait pattern deficits, with 50% of mice unable to locomote linearly, while exposure to µg largely yielded mice non-functional. In conclusion, continuous exposure to 1G of AG aboard ISS prevented gait pattern deficits and maintained performance, but exposure to lower gravitational levels (both µg and approximating Martian gravity) resulted in profound performance deficits.

Task Progress:

Neuromuscular analyses: Front paw grip strength increased in HGC mice (+22% vs pre-flight) and decreased in all AG groups (up to - 35%, p=0.0032). All groups experienced an increase in rear paw grip strength that was directly related to the level of AG, with 0G, 1G, and HGC mice experiencing a 14.9±18.3%, 37.7±14.3%, and 47.7+ 12.0% increase versus pre-flight, respectively. Gastrocnemius and soleus wet mass were related to the level of AG (p<0.0001 and p=0.0002, respectively), with HGC animals having the largest muscles and the 0G mice having the smallest. The ex vivo EIM analysis showed that animals exposed to AG had significant alterations in 50kHz parameters. An analysis of the intrinsic muscle properties also highlighted a dose-dependent reduction in both conductivity and relative permittivity (p=0.0036).

These results show that AG provided in-flight partially mitigated muscle atrophy and weakness. Impedance analysis showed that the muscle of flight animals has decreased ability to transmit electrical current (conductivity) and reduced polarizability (relative permittivity), features indicative of myofiber atrophy. These changes in the muscle's electrical properties are dose-dependent and are partially mitigated by increased levels of AG, thus emphasizing the suitability of EIM as a biomarker for microgravity-induced changes in muscle tissue composition and structure.

Bone: We found that femoral bone mineral density (BMD) was lower in 0G mice (-12.3%, p<0.0001) than HGC. AG partially mitigated the effects of microgravity as 0.33G, 0.66G, and 1G mice all had significantly greater femoral BMD than 0G mice. At the distal femur and proximal tibia, 0G mice had ~30% less trabecular bone, with 10.5% thinner and 6% fewer trabeculae than HGC (p<0.05 for all). Mice exposed to AG at the 1G level had significantly greater trabecular bone volume at the distal femur (p=0.03) and proximal tibia (p=0.007) than 0G. 1G mice also had a greater number of trabeculae than 0G mice. The 1G and HGC groups had similar trabecular bone outcomes. However, AG did not completely prevent the loss of bone in the trabecular compartment, as the mice exposed to 0.67G had less trabecular bone, with thinner and fewer trabeculae than HGC mice. Cortical area (-14.7%, p<0.001) and thickness (-9.6%, p<0.001) of the femoral midshaft were lower in 0G than HGC. Exposure to AG prevented the decrement in cortical thickness at the femoral diaphysis. Vertebral strength was lower in 0G than HGC mice (-15%, p=0.001) while AG had mixed effects on estimated vertebral strength with 1G mice similar to HGC and 0.67G lower than HGC (-13%, p=0.0025).

Histomorphometric analyses are ongoing.

These results confirm the deleterious effects of microgravity on the murine skeleton. Artificial gravity induced by centrifugal force mitigated these deleterious effects, though there was no clear dose-dependent response.

Gene Expression in Cortical Bone Spaceflight induces significant changes in gene expression in the femur (1268 DEG p<0.05): (A) 175 genes were significantly upregulated with log2fc >1, and 152 genes were significantly downregulated with log2fc < -1. (B) With 0.33G of artificial gravity, many genes are differentially expressed relative to HGC. In 0.67G, there was partial mitigation of the total number of DEG from spaceflight, with 618 DEG relative to HGC. 1G of artificial gravity almost completely prevents changes in gene expression from spaceflight, with only 4 DEG relative to HGC. (C) Gene ontology (GO) terms were padj<0.05 in upregulated genes are shown with the number of upregulated genes for each pathway denoted on the plot. No GO terms were enriched in downregulated genes. (D) The heatmap shows the 1268 genes differentially expressed at 0G relative to HGC gradually return to HGC levels with increasing levels of artificial gravity.

Single-cell RNASeq of bone marrow: Approximately 30,000 cells/group (~5,000 cells/HIVE) were obtained after quality control metrics were applied: 200-5000 genes read per cell, and <20% mitochondrial DNA mapped per cell. After aligning sequencing reads to the mouse genome (GRCm38, BeeNet) and performing initial UMAP cell clustering, we noted that 0g mice had increased percentage of cells expressing biomarkers C3 and Uhrf2 (cluster 0: 10.26% in 0g, 12.69% in 1g, 13.21% in HGC) and Tph1 and Trap1a (cluster 22: 4.43% in 0g, 0.01% in 1g, 0.02% in HGC). 1g mice had an intermediate percentage of cells expressing biomarkers Dnajc8 and Dnah8 (cluster 12: 1.93% in 0g, 2.77% in 1g, 3.69% in HGC). 0g and 1g mice had comparable percentages of cells expressing Tpm4 and Neat1, though both were lower than HGC mice (cluster 28: 0.54% in 0g, 0.48% in 1g, 0.75% in HGC). Moreover, we identified unique highly variable genes between different experimental groups (Ibsp and Gm17450 in 0g, Iglc2 in 1g, and Epx in HGC). Hbb-bs was a highly variable gene in 1g and HGC, but not 0g.

Single-cell RNA sequencing of femur bone marrow cells isolated from spaceflight mice reveals expected cellular heterogeneity. The analysis is ongoing to assign a biological function to the distinct cellular clusters and to investigate cluster-specific differential gene expression due to spaceflight ± artificial gravity. These preliminary results point to differential responses in genes involved with immune function (C3, Trap1a, Iglc2, Epx), cytoskeleton and flagellum organization (Dnah8, Tpm4), serotonin synthesis (Tph1), neuronal function (Dnajc8), erythropoiesis (Hbb-bs), and bone matrix mineralization (Ibsp). Future work will integrate the single-cell transcriptomic analysis of bone marrow with phenotypic immune, hematopoietic, and skeletal changes seen during the MHU-8 mission.

Presentations and Publications to Date Abstracts Coulombe J, Wadhwa A, Mortreux M, Emerzian S, Ciola J, Chow J, Brooks D, Willey J, Fuller C, Takahashi S, Hotz-Vitaterna M, Bouxsein ML. Response of the Murine Musculoskeletal System to Artificial Gravity Induced by Centrifugation on the International Space Station: A NASA-JAXA Collaboration. Poster presented at: The American Society of Bone and Mineral Research; Oct 13-16, 2023; Vancouver, Canada.

Wadhwa A, Coulombe J, Emerzian S, Chow J, Brooks D, Willey J, Fuller C, Takahashi S, Hotz-Vitaterna M, Bouxsein ML. Losses in murine skeletal health are partially mitigated by artificial gravity induced by centrifugation on the International Space Station: A NASA-JAXA collaboration. Presented at: The American Society of Gravitational and Space Research; Nov 14-18, 2023; Washington, D.C.

Mortreux M, Sung DM, Nagy J, Coulombe J, Ciola J, Wadhwa A, Emerzian S, Rutkove S, Fuller C, Takahashi S, Hotz-Vitaterna M, Willey J, Bouxsein ML. Artificial Gravity Induced by Centrifugation on the International Space Station Mitigates Muscle Deconditioning in Mice: the MHU-8 Mission. Presented at: The American Society of Gravitational and Space Research; Nov 14-18, 2023; Washington, D.C.

Presented at HRP-IWS in 2024 Response of the murine skeleton to artificial gravity induced by centrifugation on the International Space Station: a NASA-JAXA collaboration. J. Coulombe, A. Wadhwa, M. Mortreux, S. Emerzian, D. Brooks, J. Hanlon, J. Chow, J. Ciola, D. Hu, S. Berry, F. Gori, R. Baron, J. Willey, C. Fuller, S. Takahashi, M. Hotz-Vitaterna, M.L. Bouxsein.

	Single cell transcriptomics of femoral bone marrow from rodents exposed to artificial gravity on the International Space Station: A NASA-JAXA collaboration. A. Wadhwa, C. M. Mazur, K. Strauss, J. Coulombe, C. Fuller, S. Takahashi, M. Hotz-Vitaterna, M.L. Bouxsein, M.N. Wein. Artificial gravity during spaceflight prevents gait and performance deficits in a gravity-dose dependent manner. J.S. Willey, C.M.Patel, K.Reno, R.Y.L. Kim, J.Coulombe, C.Fuller, S. Takahashi, M.Hotz Vitaterna, M. L. Bouxsein. Arthritic responses of mouse knee menisci to artificial gravity aboard the international space station. C.M. Patel, S.S. Vander Wiele, K.E. Reno, L. Kim, J. Coulombe, C. Fuller, S. Takahashi, M. Hotz-Vitaterna, M.L. Bouxsein, J.S. Willey. Future work Additional work to be done in the no-cost extension period: 1) Complete transcriptomic analyses; 2) Complete analysis of histology and histomorphometry for both bone and muscle samples; and 3) Prepare and submit multiple manuscripts.
Bibliography Type:	Description: (Last Updated: 04/02/2026)
Articles in Peer-reviewed Journals	Patel CM, Kim L, Reno K, Coulombe J, Vitaterna MH, Fuller C, Takahashi S, Bouxsein ML, Willey JS. "Artificial gravity during spaceflight prevents gait and performance deficits in a gravity-dose dependent manner." Life Sci Space Res. 2026 Jan 16. https://doi.org/10.1016/j.lssr.2026.01.003 , Jan-2026
Articles in Peer-reviewed Journals	Tsuji R, Fujita R, Hayashi T, Sadaki S, Matsumoto T, Inoue Y, Murakami Y, Hamada M, Muratani M, Kobayashi H, Yumoto A, Okada M, Kamimura D, Okada R, Suzuki T, Kurosawa R, Otsuki A, Koshiba S, Vitaterna MH, Fuller CA, Mortreux M, Sung DM, Ciola J, Rutkove SB, Coulombe J, Kudo T, Yamamoto M, Bouxsein ML, Shiba D, Takahashi S. "0.33g mitigates muscle atrophy while 0.67g preserves muscle function and myofiber type composition in mice during spaceflight." Sci Adv. 2026 Mar 13;12(11):eaed2258. https://doi.org/10.1126/sciadv.aed2258 ; PubMed PMID: 41824581; PubMed Central PMCID: PMC12985678 , Mar-2026
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